

Selection for language-model context, read from the data

Using entropitics to decide how many retrieved items to keep, which to keep, and where foreign material sits

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Abstract

A language model's context is assembled from several sources at once: a retrieval-augmented prompt, an agent scratchpad, a long conversation. The decisions made about it — how many retrieved items to keep, which of them to trust, whether an answer stayed on its evidence — are conventionally made with one number per item against one summary of the whole: a cosine, a centroid, a fixed top- k .

This work reads the context as the K directions it resolves above its own noise floor, using the **entropitics** instrument, and makes each of those decisions from that span. K is counted from the data, so nothing is trained, no threshold is set, and no constant is fitted per corpus.

Where relevance spreads across several facets, the derived cut holds F_1 at 1.000 while a cosine cut falls to 0.000. On locating a real prompt injection planted in real documents, it names the attack as the first item to inspect three times as often as a centroid does at five sources, and leads a k-nearest-neighbour detector outright at one and two sources. On deciding whether a continuation belongs to its context, it leads a centroid by a margin that grows from one source to five.

One statement covers all of it. **These reads beat a single summary statistic exactly when the deciding distribution is multi-modal, and the margin grows with how multi-modal it is.** Where the deciding distribution has one mode they reduce to the statistic they are measured against and tie, which four BEIR corpora and an end-to-end HotpotQA run confirm.

1. The claim

A read taken over the K directions a context resolves decides better than one taken against a single summary of it, exactly when the deciding distribution is multi-modal, and the advantage grows with how multi-modal it is.

Three measurements carry it, on three different axes of modality: facets of a query's relevance (§4), sources in a context (§5), and topics spanned by a conversation (§6). Two real-corpus measurements confirm the tie where a distribution has one mode (§7).

2. A span in place of a point

Take a context $C = \{x_1, \dots, x_n\}$ of direction-normalised embeddings drawn from M distinct sources. The centroid

$$\bar{x} = \frac{1}{n} \sum_i x_i / \left\| \frac{1}{n} \sum_i x_i \right\|$$

is a single point on the sphere, and ranking by cosine to it is identical to ranking by average similarity to the rest of the set — the standard summary an item is usually scored against. Where the sources occupy M well-separated directions, $\bar{x}$ is their weighted mean and for $M > 1$ it points into the gap between them. The cosine of a legitimate item from source m then falls roughly as $\cos \theta$, with θ that item's angle from the consensus direction — a quantity set by **how spread the context is**, carrying no information about the item.

A subspace read puts a span where that point was. Let $X \in \mathbb{R}^{n \times d}$ stack the context, let $X = U\Sigma V^\top$, and take

$$V_K = [v_1, \dots, v_K], \quad K = \#\{\sigma_j : \sigma_j \text{ resolves above the noise floor}\}.$$

Score an item by the energy it retains under projection:

$$\phi_{\text{in}}(y) = \frac{\|V_K^\top y\|^2}{\|y\|^2} \in [0, 1].$$

Each resolved source direction lies in $\text{span}(V_K)$, so a legitimate item from *any* source scores high: its own direction is represented whether or not the others are. The span admits every source at once, and the margin over a single-point summary widens as the sources multiply, because the point moves further from all of them while the span keeps holding each.

3. Choosing K without choosing anything

K is the one quantity that could be tuned, and it is read rather than set. It comes from the singular spectrum via the **entropitics** instrument [1], which counts the modes standing above the noise floor derived from the block's own shape. For an $n \times d$ block of pure noise the singular values follow a Marchenko-Pastur law with a Tracy-Widom edge; modes above that edge are the ones the block resolves. The count is a property of the data and moves with the context: in §5 it reads 5.6 modes at one source and 12.5 at five, without being told how many sources there were.

The block is read **as recorded**, column means included. This read counts modes and a constant level is a mode: in the optics the library speaks it is the zero-order beam, physically present and passed by the aperture like any other. For a set of embeddings that matters directly, because **the shared mean direction is the topic the set is about** — it is the leading coherent mode, and it carries signal.

Absence is handled by masking. A row or column that was never measured is dropped before the decomposition and never reaches the denominator. A channel of measured zeros is an observation of no power and counts as a mode. The two are distinct inputs and give distinct values of K , which is what a caller with sparse features needs to know.

4. How many to keep, when relevance has several facets

The largest margin in this work is on selection. A query whose relevance is spread across R independent facets is the multi-modal case at its cleanest, and R is a controlled variable: the facets are planted, so the axis is exact rather than estimated.

Both sides run the same parameter-free lock over the same head screen; the only difference is whether the screen precedes the lock ([research/benchmarks/bench_facets.py](#), 60 items, 12 gold, 40 draws).

facets R	cosine F_1	entryptics F_1	edge	95% CI
1	1.000	1.000	+0.000	[+0.000, +0.000]
2	1.000	1.000	+0.000	[+0.000, +0.000]
3	1.000	1.000	+0.000	[+0.000, +0.000]
4	0.806	1.000	+ 0.194	[+0.099, +0.301]
5	0.000	1.000	+ 1.000	[+1.000, +1.000]
6	0.000	1.000	+ 1.000	[+1.000, +1.000]

Entryptics holds F_1 at 1.000 at every facet count. The cosine cut matches it while the relevance has one to three facets and falls to 0.000 by five. The reason is §2's, transposed from a context to a query: one scalar carries one facet's worth of evidence, and each gold item's total weakens toward the distractor level as the facets multiply, where an R -mode subspace holds all of them at once.

The count kept is derived per query throughout. Nothing is swept, and the same lock produces both columns.

5. Which item is foreign, when a context spans several sources

The second axis of modality is the number of sources a context is assembled from. Legitimate content is whole wikitext-103 articles, so each source is a genuine topic; the attacks are the [deepset/prompt-injections](#) corpus, strings written to attack real systems. One attack is appended to a context built from M sources and the read ranks it. Encoder is [bge-small-en-v1.5](#), 150 trials per row, intervals paired per trial ([research/benchmarks/bench_injection.py](#)).

Two baselines. The **centroid** is the standard single summary of §2. **k-NN** scores an item by its similarity to its five nearest neighbours — a local read, so it carries no penalty for a context that spans topics, and it is the stronger of the two.

sources	modes	items	entryptics	centroid	k-NN	vs centroid	vs k-NN
1	5.6	39	0.990	0.983	0.983	+0.007 [+0.003, +0.012]	+ 0.007 [+0.003, +0.011]
2	7.5	78	0.970	0.901	0.960	+0.069 [+0.052, +0.088]	+ 0.010 [+0.004, +0.017]
3	9.4	115	0.941	0.811	0.940	+0.131 [+0.105, +0.157]	+0.001 [−0.007, +0.010]
5	12.5	192	0.907	0.733	0.927	+0.174 [+0.147, +0.201]	−0.020 [−0.031, −0.010]

Against the standard summary the margin grows from +0.007 to +0.174, every interval excluding zero, exactly as §2 predicts: the centroid falls from 0.983 to 0.733 as the sources multiply while entryptics holds

from 0.990 to 0.907.

Against a local detector entropitics leads where the context is tightest — +0.007 and +0.010 at one and two sources, both intervals excluding zero — and the two converge as the context widens.

The operator's question is which item to open first, and there entropitics leads at every source count:

sources	entropitics	centroid	k-NN
1	0.787	0.740	0.707
2	0.467	0.253	0.380
3	0.287	0.140	0.267
5	0.160	0.053	0.160

At five sources it names the attack first three times as often as the centroid does, from a pool of 192 items, and matches the local detector. Each item is scored against the subspace of the set without it; Appendix A gives the reason.

6. Whether a continuation belongs to its context

The third axis is a conversation's topic spread, and the question is non-adversarial: **is this continuation on-topic?** Held-out sentences from the context's own sources — on-topic, and held out of the context — are scored against sentences from a different article. Same encoder, 150 trials per row, six held-out sentences per source so the target count resolves every effect reported ([research/benchmarks/bench_drift.py](#)).

sources	modes	context	entropitics	centroid	k-NN	vs centroid
1	2.0	12	0.955	0.954	0.960	+0.001 [−0.002, +0.004]
2	2.8	24	0.931	0.893	0.945	+ 0.038 [+0.031, +0.046]
3	3.8	36	0.922	0.838	0.942	+ 0.084 [+0.074, +0.095]
5	5.8	60	0.872	0.717	0.909	+ 0.155 [+0.141, +0.169]

The same curve, from an independent task. The edge over the standard summary opens at two sources and grows to +0.155 at five, every interval from two upward excluding zero, with the centroid falling 0.954 → 0.717 while entropitics holds 0.955 → 0.872. §5 appends foreign text and this withholds native text; they were measured separately and agree on where the crossover sits and on how the margin grows.

A local neighbour read leads on this task throughout, by 0.005 to 0.038. Ranking a target against its nearest context sentences is a close fit to the question as posed, and the value entropitics adds here is the count it derives rather than the ordering it produces — §4 and §7 measure that directly.

7. Real corpora, where relevance has one mode

The law predicts a tie wherever the deciding distribution has one mode, because the read reduces to the statistic it is measured against. Two real-corpus measurements confirm it.

Four BEIR corpora — nfcopus, scifact, arguana, fiqa, 2,421 queries — against the identical `top_break` lock run on plain cosine salience, so the screen is the only difference ([research/benchmarks/bench_beir.py](#)):

dataset	cosine + lock	entropitics	edge	95% CI
nfcopus	0.295	0.292	−0.003	[−0.017, +0.010]
scifact	0.571	0.557	−0.013	[−0.029, +0.001]
arguana	0.257	0.255	−0.002	[−0.008, +0.004]
fiqa	0.344	0.342	−0.002	[−0.012, +0.007]

Four ties, every interval containing zero. At the point the cut sees them, 12.2% of the 3,265 pooled queries carry four or more relevant documents in their horizon, so the deciding distribution has one mode and \$4's condition is absent.

HotpotQA end-to-end, 150 questions, a language model answering from the selected context ([research/benchmarks/bench_hotpot_e2e.py](#)). The decisive arm is cosine's top- m , where m is the count entropitics derived *for that question*, so the only difference is which items were chosen:

model	fixed $k = 5$	cosine top- m	entropitics	entropitics − top- m
Qwen2.5-1.5B	0.487	0.436	0.436	+0.000 [+0.000, +0.000]
Qwen2.5-7B	0.676	0.545	0.543	−0.001 [−0.020, +0.019]

The selection matches cosine's at both model sizes, as the law requires on unimodal relevance.

What the derived count buys where the reads tie is the count itself. The best single constant k gives up as much as 0.086 F_1 against a corpus's own tuned k ; reading the count per query gives up at most 0.019. That is the value where there is no sweep to run — a service answering against more than one corpus, or a corpus whose content moves.

Size the budget to the task. On HotpotQA the derived count spends 36% of a fixed top-5's context tokens, keeping 2.1 paragraphs where a multi-hop answer needs two gold paragraphs. Against that fixed top-5 the difference is −0.133 [−0.202, −0.067] at Qwen2.5-7B, and it is the count rather than the ranking: the matched-budget arm above shows the ranking already matches the baseline. Where a task needs several passages, set a floor on the count.

8. What this costs

One thin SVD of the context block, $O(nd \min(n, d))$, on embeddings the pipeline has already computed for retrieval. No additional model call, no second encoder, no index. Scoring an item against the basis is a $d \times K$ matrix-vector product. For the context sizes here — tens to a few hundred items — the read is well under the cost of the retrieval that produced them.

The anomaly read of \$5 scores each item against the subspace of the set without it, which is n decompositions rather than one. Appendix A gives the reason and the alternative.

9. Scope

Coherence reads topic. An answer that stays on its sources' subject scores high whether or not it is correct: on the 150 HotpotQA questions, coherence separated right answers from wrong ones by +0.008 at 1.5B and +0.010 at 7B. It is a monitor for off-source and lost-thread behaviour.

Scores and rankings. The reads order items within one context. "Is there an anomaly in this set at all" is a question about a reference distribution, which a caller supplies; `anomaly_z` takes one.

Measured on one encoder. All embedding numbers are `bge-small-en-v1.5`, and the end-to-end tables are Qwen2.5-1.5B and Qwen2.5-7B. The mechanism in §2 is a statement about direction-normalised embeddings generally; the magnitudes are measured on these models, and K depends on how a model spreads a topic.

10. Reproducing

```
pip install -e ".[bench]"
python -m research.benchmarks.bench_facets           # §4
python -m research.benchmarks.bench_injection        # §5
python -m research.benchmarks.bench_drift            # §6
python -m research.benchmarks.bench_beir            # §7
```

The end-to-end table in §7 needs a language model as well:

```
pip install -e ".[bench,bench-llm]"
python -m research.benchmarks.bench_hotpot_e2e --model Qwen/Qwen2.5-7B-Instruct --n 150
```

`bench_facets` runs on CPU in seconds and needs no corpus. The others download their data from Hugging Face. Every number in this paper carries the conditions it was measured under.

Appendix A. Two measurements the API's shape rests on

A.1 — Each item is scored against the subspace of the set without it. `anomaly` leaves the item out before building the span. Scoring an item against a span it helped define gives the item its own coherent mode once it is foreign enough to clear the noise floor, at which point it reads as maximally normal. One injected item in a clean multi-topic set, 15 draws, top-1 localisation:

injected item is	read	1 topic	3 topics	5 topics
70% off the topics	either	1.00	1.00	1.00
90% off	in-set	1.00	0.80	1.00
90% off	leave-one-out	1.00	1.00	1.00
fully orthogonal	in-set	1.00	0.07	1.00
fully orthogonal	leave-one-out	1.00	1.00	1.00

The effect moves with whether the resolved rank reaches the outlier's own mode, which a caller cannot predict from anything visible to them, so leave-one-out is the default: it scored 1.00 in every cell. The in-set read is available for latency and is sound where the set holds no hard outlier, which is the ordinary case for text — real injections land inside the embedder's cone and a partial outlier does not earn a mode.

A.2 — A subspace overlap is read against its null. In high dimension two subspaces that are not specifically aligned are near-orthogonal, so a raw `subspace_overlap` reports "most of the structure lies outside" for a random matrix as readily as for a real one, measuring the ambient dimension. The analytic null is k_b/d — a random k_b -dimensional subspace covers that fraction of anything — and `chance_overlap` returns it. Report the ratio of measured overlap to chance.

References

- [1] **entryptics** — the spectral instrument. Resolved-mode counting against a Marchenko-Pastur/Tracy-Widom noise floor, and the occupancy read used here to set K . Apache-2.0. <https://pypi.org/project/entryptics/>
- [2] **wikitext-103** — Merity et al., *Pointer Sentinel Mixture Models*, ICLR 2017.
- [3] **deepset/prompt-injections** — a corpus of real prompt-injection strings.
<https://huggingface.co/datasets/deepset/prompt-injections>
- [4] **BEIR** — Thakur et al., *BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models*, NeurIPS 2021.
- [5] **HotpotQA** — Yang et al., *HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering*, EMNLP 2018.

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